

EDUCATION	University of Wisconsin Madison	Madison, Wisconsin, US
	<i>Ph.D., Environmental Chemistry and Technology</i>	Sep 2022 - Expected Jun 2025
	Research Interest: Optical sensors for pesticide detection in environmental systems.	
	Beijing Normal University	Beijing, China
	<i>Master of Engineer, Environmental Science</i>	Sep 2019 - Jun 2022
	Research Interest: Microplastics transport and fate in aquatic environments.	
	Chongqing University	Chongqing, China
	<i>Bachelor of Engineer, Environmental Science</i>	Sep 2015 - Jun 2019
	Research Interest: Air pollution characterization, with a focus on PM_{2.5}.	
SELECTED PUBLICATIONS	1. Liu, S., & Wei, H. (2025). Enhanced ligand exchange for ultrasensitive neonicotinoid detection in soil water via surface-enhanced Raman spectroscopy. <i>Water Research</i> , 286, 124268.	
	2. Wan, S., Chae, H., Meese, A. F., Nwokonkwo, O., Arrazolo, L., Yip, K. L., Ma, X., Liu, S., Muhich, C., Wang, D., Wei, H., & Kim, J. H. (2025). Overcoming the reactivity-stability challenge in water treatment catalyst through spatial confinement. <i>Nature Communications</i> , Accepted.	
	3. Liu, S., Lazarcik, J., & Wei, H. (2024). Emerging investigator series: quantitative insights into the relationship between the concentrations and SERS intensities of neonicotinoids in water. <i>Environmental Science: Nano</i> , 11(8), 3294-3300.	
	4. Ho, D., Liu, S., Wei, H., & Karthikeyan, K.G. (2024). The glowing potential of Nile red for microplastics Identification: Science and mechanism of fluorescence staining. <i>Microchemical Journal</i> , 197, 109708.	
	5. Liu, S., Shang, E., Liu, J., Bolan, N.S., Kirkham, M.B., & Li, Y. (2022). What have we known so far for fluorescence staining and quantification of microplastics: A tutorial review. <i>Frontiers of Environmental Science & Engineering</i> , 16(1), 8.	
	6. Li, Y., Liu, Y., Liu, S., Zhang, L., Shao, H., & Zhang, W. (2022). Leaching Mechanisms of Bisphenol S from Polyethersulfone (PES) and Polyphenylsulfone (PPSU) Microplastics During Photoaging Processes. <i>Environmental Science & Technology</i> , 56(5), 3033-3044.	
PROJECT EXPERIENCE	U.S. Environmental Protection Agency Science to Achieve Results (STAR) Program	Jun 2023 - April 2025
	- Led the development of optical sensors for ultrasensitive neonicotinoid detection, leveraging surface-enhanced Raman spectroscopy (SERS). - Collaborated with researchers from the University of Florida on biochar concentration modules and Yale University on advanced oxidation processes (AOP) for integrated sensor systems to remove neonicotinoids from rural rivers.	

TEACHING & MENTORING	• CEE 325 Environmental Engineering Materials, Lab Session – Instructed and assisted 50 undergraduate students in a lab setting. – Provided detailed instrument instructions and hands-on lab support. – Supported undergraduates' career development by serving as a professional reference for lab-position applications. • DELTA Mentoring Program – Completed comprehensive training and earned a certified mentoring badge. – Mentored 4 undergraduate researchers, providing project design guidance, hands-on supervision, and fostering independent research skills.	
AWARDS AND HONORS	• National Scholarship Award	2016
	• National Scholarship Award	2017
	• Outstanding Graduate of Chongqing University	2019
	• Excellent Award for Graduation Paper of bachelor degree	2019
	• First Class Scholarship for Postgraduate of Beijing Normal University	2020
SKILLS	Instrumentation: Raman spectroscopy, High-Performance Liquid Chromatography (HPLC), Scanning Electron Microscopy (SEM), Dynamic Light Scattering (DLS).	
	Programming Language: Python.	
ACADEMIC SERVICES	Community Outreach: Provided scientific insights on neonicotinoid detection in the Black Earth Creek watershed to a community partner.	